

# Homework 2

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October 15, 2025

## Exercise 1

Let  $a, b, c \in \mathbb{Z}$  such that  $(a, b) = (a, c) = 1$ . Suppose  $(a, bc) = n$  where  $n > 1$ . Let  $p$  be the largest prime factor of  $n$ . Since  $n \mid a$  and  $p \mid n$ ,  $p \mid a$ . Likewise,  $p \mid bc$ . Consequently, either  $p \mid b$  or  $p \mid c$ . If  $p \mid b$ , then  $(a, b) \neq 1$ , since  $p > 1$  divides both  $a$  and  $b$ . Similarly, if  $p \mid c$ ,  $(a, c) \neq 1$ . In either case, this is a contradiction; therefore  $(a, bc) = 1$ .

## Exercise 2

### Part (a)

Let  $a, b, c \in \mathbb{Z}$  such that  $(a, b) = 1$ ,  $a \mid c$ , and  $b \mid c$ . By Bezout's theorem, there exist  $m, n \in \mathbb{Z}$  such that  $ma + nb = 1$ . Then:

$$\begin{aligned}ma + nb &= 1 \\mac + nbc &= c\end{aligned}$$

If  $a \mid c$ , there exists  $p \in \mathbb{Z}$  such that  $ap = c$ . Likewise, there exists some  $q \in \mathbb{Z}$  such that  $bq = c$ . Then:

$$\begin{aligned}ma + nb &= 1 \\ma(bq) + nb(ap) &= c \\ab(mq + np) &= c\end{aligned}$$

Consequently,  $ab \mid c$ .

However, this is not true if  $(a, b) \neq 1$ . Note the case where  $a = 6$ ,  $b = 9$ , and  $c = 18$ . Clearly,  $a \mid c$  and  $b \mid c$ , but  $ab \nmid c$ .

### Part (b)

Let  $a, b, c \in \mathbb{Z}$  such that  $(a, b) = 1$  and  $a \mid bc$ . By Bezout's theorem, there exist  $m, n \in \mathbb{Z}$  such that  $ma + nb = 1$ . Then:

$$\begin{aligned}ma + nb &= 1 \\mac + nbc &= c\end{aligned}$$

If  $a \mid c$ , there exists  $p \in \mathbb{Z}$  such that  $ap = c$ . Then:

$$\begin{aligned}ma + nb &= 1 \\ma + nb(ap) &= c \\a(m + nbp) &= c\end{aligned}$$

Consequently,  $a \mid c$ .

However, this is not true if  $(a, b) \neq 1$ . Note the case where  $a = 6$ ,  $b = 3$ , and  $c = 4$ . Clearly,  $a \mid bc$ , but  $a \nmid c$ .

### Exercise 3

#### Part (a)

Applying the division algorithm repeatedly:

$$1234 = 981 \cdot 1 + 253$$

$$981 = 253 \cdot 3 + 222$$

$$253 = 222 \cdot 1 + 31$$

$$222 = 31 \cdot 7 + 5$$

$$31 = 5 \cdot 6 + 1$$

$$5 = 1 \cdot 5 + 0$$

Consequently,  $(981, 1234) = 1$ . Expressing the GCD as a linear combination:

$$\begin{aligned} 1 &= 5 \cdot -6 + 31 \\ &= (31 \cdot -7 + 222) \cdot -6 + 31 = 31 \cdot 43 + 222 \cdot -6 \\ &= (222 \cdot -1 + 253) \cdot 43 + 222 \cdot -6 = 222 \cdot -49 + 253 \cdot 43 \\ &= (253 \cdot -3 + 981) \cdot -49 + 253 \cdot 43 = 253 \cdot 190 + 981 \cdot -49 \\ &= (981 \cdot -1 + 1234) \cdot 190 + 981 \cdot -49 = 981 \cdot -239 + 1234 \cdot 190 \end{aligned}$$

The GCD can be expressed as  $1 = 981 \cdot -239 + 1234 \cdot 190$ .

#### Part (b)

Applying the division algorithm repeatedly:

$$100313 = 34709 \cdot 2 + 30895$$

$$34709 = 30895 \cdot 1 + 3814$$

$$30895 = 3814 \cdot 8 + 383$$

$$3814 = 383 \cdot 9 + 367$$

$$383 = 367 \cdot 1 + 16$$

$$367 = 16 \cdot 22 + 15$$

$$16 = 15 \cdot 1 + 1$$

$$15 = 1 \cdot 15 + 0$$

Consequently,  $(34709, 100313) = 1$ . Expressing the GCD as a linear combination:

$$\begin{aligned}
 1 &= 15 \cdot -1 + 16 \\
 &= (16 \cdot -22 + 367) \cdot -1 + 16 = 16 \cdot 23 + 367 \cdot -1 \\
 &= (367 \cdot -1 + 383) \cdot 23 + 367 \cdot -1 = 367 \cdot -24 + 383 \cdot 23 \\
 &= (383 \cdot -9 + 3814) \cdot -24 + 383 \cdot 23 = 383 \cdot 239 + 3814 \cdot -24 \\
 &= (3814 \cdot -8 + 30895) \cdot 239 + 3814 \cdot -24 = 3814 \cdot -1936 + 30895 \cdot 239 \\
 &= (30895 \cdot -1 + 34709) \cdot -1936 + 30895 \cdot 239 = 30895 \cdot 2175 + 34709 \cdot -1936 \\
 &= (34709 \cdot -2 + 100313) \cdot 2175 + 34709 \cdot -1936 = 34709 \cdot -2414 + 100313 \cdot 2175
 \end{aligned}$$

The GCD can be expressed as  $1 = 34709 \cdot -2414 + 100313 \cdot 2175$ .

## Exercise 4

Let  $n \in \mathbb{N}$ . Since  $(2n^2 + 6n - 4) - (2n^2 + 4n - 3) = 2n - 1 > 0$ ,  $2n^2 + 6n - 4 > 2n^2 + 4n - 3$ . Applying the division algorithm repeatedly:

$$2n^2 + 6n - 4 = (2n^2 + 4n - 3) \cdot 1 + (2n - 1)$$

$$2n^2 + 4n - 3 = (2n - 1) \cdot (n + 2) + (n - 1)$$

$$2n - 1 = (n - 1) \cdot 2 + 1$$

$$n - 1 = 1 \cdot (n - 1) + 0$$

Consequently,  $(2n^2 + 6n - 4, 2n^2 + 4n - 3) = 1$ .